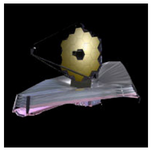

8872 - The Dark Side of the Force: Unraveling Protostellar Jet Asymmetry by Probing TMC1A's Fainter Red-shifted Outflow with JWST

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
TMC1A Redshifted Outflows				
	1	NIRSpec IFU	NIRSpec IFU Spectroscopy	(1) TMC1A-NIRSpec
	2	MIRI MRS	MIRI Medium Resolution Spectroscopy	(2) TMC1A-MIRI
	3	MIRI Background	MIRI Medium Resolution Spectroscopy	(3) Taurus-Bkg

ABSTRACT

We propose to observe the red-shifted atomic jet and molecular outflow of Class I protostar TMC1A using JWST's NIRSpec and MIRI MRS. Archival programs (PID: 1290,2104) have mapped the blue-shifted outflow, revealing a collimated red-shifted [Fe II] jet and wide-angle H2 wind with a unique asymmetry: both [Fe II] and H2 outflows are fainter on the red-shifted side, but the H2 outflow is over 3 times brighter than the [Fe II] jet relative to their blue-shifted counterparts.

This could suggest that the relative strength of H2 outflow and [Fe II] jet could be different between the red- and blue-shifted side, indicating that the outflow and jet are driven independently from different parts of the disk. As the red-shifted emission is at the edge of the archival data's FOV, we propose observations centered on this component. By obtaining a complete map of the red-shifted jet/outflow, we aim to study differences in bipolar jet+outflow morphology and physical properties (e.g., density, temperature, mass outflow rates) between the red- and blue-shifted sides. This will help distinguish between intrinsic (e.g., differential mass loss) and extrinsic (e.g., ambient medium interactions, extinction) explanations for the observed bipolar asymmetries. Leveraging JWST's sensitivity to probe this deeply embedded system, our study will offer insights into protostellar outflow asymmetries and help constrain launching mechanisms. These findings have implications beyond star formation, as the same launching mechanisms and similar asymmetries in jets/outflows are observed around a wide range of astrophysical systems.

OBSERVING DESCRIPTION

We propose to observe the red-shifted atomic jet and molecular outflow of the Class I protostar TMC1A using JWST's NIRSpec IFU and MIRI MRS instruments. These observations will complement existing public data (proposal IDs: 2104, 1290) that primarily mapped the blue-shifted outflow.

NIRSpec IFU observations will cover 1-5 microns using G140H, G235H, and G395H gratings at R~2700, while MIRI MRS will cover 5-28 microns across all channels. This complete wavelength coverage is crucial for detecting key [Fe II] and H2 lines needed for our extinction and excitation analyses.

Our pointing is offset independently for NIRSpec and MIRI based previous observations to focus on the red-shifted outflow while maintaining partial overlap with existing datasets for alignment. No target acquisition is planned as we're observing off-protostar.

We maintain similar exposure setups to the previous proposals. For MIRI, we double the number of integrations per exposure in this proposal to boost the S/N of key H2 lines, while maintaining low groupings per integration to prevent saturation. Line strengths for ETC calculations are based on MAST level 3 products for [Fe II] and H2 lines detected in the red-shifted outflow where available. For MIRI channels 1-3, where red-shifted outflow data is unavailable due to FOV limitations, we estimate line strengths at 5\% of the blue-shifted data brightness at similar spatial offsets. Our

ETC v4.0 calculations indicate >3 sigma detection of lines required for extinction and excitation analyses.

We use the NRSRAPID readout pattern for NIRSpec to prevent saturation. We chose the F770W filter to avoid PSF overlap in longer wavelengths, and its simultaneous imaging improves astrometric precision.

This observing strategy will provide a comparative view of both blue- and red-shifted jets and outflows, enabling a thorough analysis of the bipolar structure and a determination of whether the asymmetries in TMC1A's jet and wider-angle outflow are intrinsic or due to external factors (such as different levels of extinction). The high sensitivity of JWST is critical for detecting and spatially resolving faint emission features in this deeply embedded system.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TMC1A-NIRSpec	RA: 04 39 35.2157 (69.8967321d) Dec: +25 41 43.73 (25.69548d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Epoch of Position: 2000	
Comments: Shifted 0.3 arcsecs to the East in RA and 1 arcsec south in Dec from Cycle 2 NIRSpec IFU observations (2104, 04 39 35.1936, +25 41 44.73) *Category=Star* *Description=[Protostars]* *Extended=YES*				
(2)	TMC1A-MIRI	RA: 04 39 35.1700 (69.8965417d) Dec: +25 41 43.17 (25.69533d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Epoch of Position: 2000	
Comments: Shifted 2 arcsec south in Dec from Cycle 1 MIRI MRS observations (1290, 04 39 35.2030 +25 41 45.17) *Category=Star* *Description=[Protostars]* *Extended=YES*				
(3)	Taurus-Bkg	RA: 04 39 32.2900 (69.8845417d) Dec: +26 05 14.50 (26.08736d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Epoch of Position: 2000	
Comments: Same background location as done in proposal ID: 1290 (termed DARK-TAU) *Category=Calibration* *Description=[Telescope/sky background]*				

Observation

Proposal 8872, Observation 1: NIRSpec IFU

Thu Nov 13 22:00:48 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TMC1A-NIRSpec	RA: 04 39 35.2157 (69.8967321d)	Proper Motion RA: 0 mas/yr	
		Dec: +25 41 43.73 (25.69548d)	Proper Motion Dec: 0 mas/yr	
		Equinox: J2000	Epoch of Position: 2000	

Comments: Shifted 0.3 arcsecs to the East in RA and 1 arcsec south in Dec from Cycle 2 NIRSpec IFU observations (2104, 04 39 35.1936, +25 41 44.73)
Category=Star
Description=[Protostars]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	MEDIUM	6	6	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSRAPID	60	2	false	true	NONE	6	12	7859.316	
2	G235H/F170LP	NRSRAPID	15	2	false	true	NONE	6	12	2061.46	
3	G395H/F290LP	NRSRAPID	4	4	false	true	NONE	6	24	1288.412	

Observation

Proposal 8872, Observation 2: MIRI MRS

Thu Nov 13 22:00:48 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	TMC1A-MIRI	RA: 04 39 35.1700 (69.8965417d)	Proper Motion RA: 0 mas/yr	
		Dec: +25 41 43.17 (25.69533d)	Proper Motion Dec: 0 mas/yr	
		Equinox: J2000	Epoch of Position: 2000	

Comments: Shifted 2 arcsec south in Dec from Cycle 1 MIRI MRS observations (1290, 04 39 35.2030 +25 41 45.17)
Category=Star
Description=[Protostars]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	9	8	1	Dither 1	4	32	876.913	
1	SHORT(A)	MRSLONG		FASTR1	9	8	1	Dither 1	4	32	876.913	
1	SHORT(A)	MRSSHORT		FASTR1	9	8	1	Dither 1	4	32	876.913	
2		IMAGER	F770W	FASTR1	9	8	1	Dither 1	4	32	876.913	
2	MEDIUM(B)	MRSLONG		FASTR1	9	8	1	Dither 1	4	32	876.913	
2	MEDIUM(B)	MRSSHORT		FASTR1	9	8	1	Dither 1	4	32	876.913	
3		IMAGER	F770W	FASTR1	9	8	1	Dither 1	4	32	876.913	
3	LONG(C)	MRSLONG		FASTR1	9	8	1	Dither 1	4	32	876.913	
3	LONG(C)	MRSSHORT		FASTR1	9	8	1	Dither 1	4	32	876.913	

Special Requirements	Group Observations 2, 3, Non-interruptible

Observation

Proposal 8872, Observation 3: MIRI Background

Thu Nov 13 22:00:48 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	Taurus-Bkg	RA: 04 39 32.2900 (69.8845417d)	Proper Motion RA: 0 mas/yr	
		Dec: +26 05 14.50 (26.08736d)	Proper Motion Dec: 0 mas/yr	
		Equinox: J2000	Epoch of Position: 2000	

Comments: Same background location as done in proposal ID: 1290 (termed DARK-TAU)
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	9	8	1	None	1	8	219.228	
1	SHORT(A)	MRSLONG		FASTR1	9	8	1	None	1	8	219.228	
1	SHORT(A)	MRSSHORT		FASTR1	9	8	1	None	1	8	219.228	
2		IMAGER	F770W	FASTR1	9	8	1	None	1	8	219.228	
2	MEDIUM(B)	MRSLONG		FASTR1	9	8	1	None	1	8	219.228	
2	MEDIUM(B)	MRSSHORT		FASTR1	9	8	1	None	1	8	219.228	
3		IMAGER	F770W	FASTR1	9	8	1	None	1	8	219.228	
3	LONG(C)	MRSLONG		FASTR1	9	8	1	None	1	8	219.228	
3	LONG(C)	MRSSHORT		FASTR1	9	8	1	None	1	8	219.228	

Special Requirements	
	Group Observations 2, 3, Non-interruptible